

Sanket Uday Mane

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Professional Summary

AI/ML Engineer with hands-on experience in deep learning, computer vision, LiDAR classification, geospatial AI, and Generative AI. Skilled in implementing transformer/GPT architectures from scratch in PyTorch, fine-tuning open-source LLMs, and building RAG-based agentic AI systems with LangChain. Experienced in solving complex engineering problems and improving developer productivity through agentic AI, custom GPTs, prompt engineering, and AI automation.

Skills

- **Core:** Python, C++, SQL/MySQL, Machine Learning, Deep Learning, GenAI, Data Preprocessing & Annotation, Data Analysis
- **Tools:** PyTorch, scikit-learn, NumPy, Pandas, Matplotlib, Seaborn, Flask, Git/GitHub, VS Code, Jupyter Notebook, LangChain, YOLOv8/YOLOv12, Image Processing
- **GenAI & LLMs:** Transformer/GPT architecture from scratch (PyTorch), LLM fine-tuning, RAG, Agentic AI (LangChain), Prompt Engineering, Custom Chatbots
- **AI Process & Debugging:** AI-assisted debugging, reverse engineering, crash analysis, silent failure diagnosis, memory leak troubleshooting, AI automation

Work Experience

NakshaTech Pvt Ltd, Bangalore, Karnataka | AI/ML Engineer | Oct 2025 – Present

- Develop and train deep learning models for detection and segmentation on satellite/drone imagery and LiDAR data, handling large-scale data preprocessing and annotation for GeoAI workflows.
- Fine-tuned open-source LLMs for backend automation and a custom internal chatbot, and built RAG-based agentic AI workflows using LangChain and prompt engineering.
- Accelerate development speed and issue resolution using AI-assisted debugging, AI data analytics, custom GPTs, and AI automation; perform structured R&D before every new initiative to reliably deliver quality results on unfamiliar tasks.

PHN Technology Pvt Ltd, Pune, Maharashtra | Machine Learning Engineer Intern | Oct 2023 – Mar 2024

- Assisted in developing and training machine learning models using supervised, unsupervised, and semi-supervised learning techniques.
- Improved experimentation and troubleshooting workflows through practical use of AI tools for analysis, validation, and faster iteration.

Q-connect Business Solution, Pune, Maharashtra | Technical Support Executive (System Engineer) | Dec 2022 – Jun 2023

Education

- **SPPU, Pune, Maharashtra | B.E. in Information Technology | Sept 2021 – Mar 2024 | CGPA: 7.46/10**
Relevant Coursework: Data Structures and Algorithms, Databases, Operating Systems, Computer Networks, Machine Learning, Deep Learning, Data Science, Image Processing

Project Work

- **PointNet++ Architecture for LiDAR Data Classification:** Built an end-to-end pipeline for semantic classification of 3D LiDAR point clouds (.laz/.las) with custom PointNet++ and advanced feature extraction using Cloth Simulation Filter-based height metrics.
- **NakshAI_LiDAR Software:** Developed geospatial software for classification of geological entities as an alternative to Bentley MicroStation, integrating 6+ AI models with 95%+ accuracy to speed up classification and reduce manual effort.
- **YOLOv8/YOLOv12 Object Detection for Drone & Dashcam NH4 Road Survey:** Developed an object detection solution for vehicles, road damage, buffer zones, road cracks, and road-line condition analysis.
- **Traffic Density Analysis for Bangalore City:** Developed a traffic density dashboard for class-wise vehicle detection and time-based density estimation using a model trained on 10k Indian vehicle samples with 96%+ accuracy.

Awards and Certificates

- **CODE CATELYST Award** – Recognized on the company's 14th Anniversary Day at NakshaTech Pvt Ltd for contributions and performance as an AI/ML Engineer
- **Mentor at College:** Helping junior students and working professionals to get better at problem solving, coding, Machine Learning and Deep Learning